

REPORT: CONSTRUCTION OF GUITAR USING SOLIDWORKS

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TITLE: Construction of Guitar using SolidWorks-This project involves designing and modelling a fully detailed guitar using SolidWorks 3D CAD software.

OBJECTIVE: The objective of this project is to create a complete 3D solid model of a guitar, including individual part modelling, assembly, rendering, engineering drawings, and bill of materials (BOM). The goal is to understand CAD modelling workflows, parametric design, and technical drawing preparation.

SHORT DESCRIPTION: The guitar was modelled by creating each part separately, such as the body, neck, fretboard, tuning pegs, bridge, sound hole rosette, and pickguard. These parts were then assembled into a complete model using appropriate mates. Engineering drawings were generated to show dimensions, orthographic views, and material specifications. A bill of materials (BOM) was included to summarize all components with their assigned materials such as oak and mahogany.

Software Used: SolidWorks 2021

REFERENCES:

1. SolidWorks Official Documentation
2. YouTube tutorials for surfacing and part modelling <https://youtu.be/XQjCjsMa6jI>
3. Online references for guitar dimensions and proportions